

José Daniel Sarmiento Blanco

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PROFESSIONAL SUMMARY

Results-driven Artificial Intelligence Engineer and Software Developer with 3+ years of experience engineering high-throughput Machine Learning, Computer Vision, and Natural Language Processing (NLP) systems. Proven track record optimizing deep learning models in PyTorch and CUDA (reducing latency by 40%), building production conversational AI engines, and scaling distributed microservices in Rust and Python. Expert in translating business requirements into resilient, production-ready AI solutions supported by rigorous CI/CD and cloud-native architecture.

PROFESSIONAL EXPERIENCE

Worldpanel by Numerator

AI and Software Engineer

Remote

Dec 2025 – Present

- Architected and deployed automated data mining pipelines using **Python, .NET, and SQL**, ingesting [e.g., **500K+**] consumer market research records daily.
- Engineered real-time ETL aggregation workflows, reducing data ingestion latency by [e.g., **35%**] and accelerating analytics query responsiveness.
- Optimized complex SQL Server queries and data indexing, cutting backend reporting runtimes by [e.g., **40%**] for enterprise dashboards.

AXD.SAS

AI and Software Developer (R&D) (3+ Years Progressive Experience)

Remote

May 2022 – Sep 2025

Developer – Lab X (Multimodal Generative AI)

Aug 2025 – Sep 2025

- Spearheaded applied research in multimodal generative AI synthesizing language, vision, and speech representations, accelerating prototype iteration cycles by [e.g., **30%**].
- Constructed bilingual (Spanish/English) conversational agents with NLP, ASR, and TTS pipelines, delivering [e.g., **<250ms**] end-to-end response latency.

Software Developer (R&D) – Advanced Role

Feb 2025 – Aug 2025

- Designed and deployed computer vision pipelines with **YOLOv8, PyTorch, and TensorRT**, achieving a **40% reduction in inference latency** in production.
- Fine-tuned Transformer architectures for NLP dialogue and speech synthesis, boosting semantic intent classification accuracy by [e.g., **18%**].
- Engineered event-driven distributed streaming microservices using **Apache Kafka and RabbitMQ**, processing [e.g., **10K+ msg/sec**] reliably.

Software Developer (R&D) – Core Role

May 2022 – Feb 2025

- Developed high-concurrency microservice APIs in **Rust (Actix-web)** and **Python** to serve ML models, cutting p99 response times to [e.g., **<50ms**].
- Built production conversational analytics engines using **Transformers, spaCy, and NLTK**, parsing dialogue logs to extract actionable business insights.
- Implemented an end-to-end voice conversation pipeline (STT, NLP dialogue management, TTS) supporting [e.g., **50K+**] monthly interactive user sessions.
- Optimized ML algorithms to achieve a **300% performance improvement**; built CI/CD deployment pipelines, cutting developer onboarding by **25%**.

TECHNICAL PROJECTS

YOLOv10 C++ Real-Time Inference Engine | C++, OpenCV 4.9, ONNX Runtime 1.17, TensorRT

[GitHub Code](#)

- Developed a modular C++ computer vision application for real-time object detection using YOLOv10 and ONNX Runtime 1.17, achieving [e.g., **60+ FPS**] inference.
- Implemented Top-K layer optimizations and custom image tensor manipulation in OpenCV 4.9, reducing CPU/GPU memory footprint by [e.g., **30%**].

CIFAR-10 Deep Learning Vision Classifier | Python, TensorFlow, Scikit-learn, NumPy

[GitHub Code](#)

- Implemented a Deep Convolutional Neural Network (CNN) in TensorFlow with custom data augmentation, attaining [e.g., **91.5%**] test accuracy.
- Evaluated classification robustness across key validation metrics (Recall, AUC-ROC, Categorical Accuracy) and published reproducible benchmarks.

EDUCATION

Pontifícia Universidade Católica do Rio de Janeiro (PUC-Rio)

Rio de Janeiro, Brazil

M.S. in Electrical Engineering – Decision Support Methods (Coursework: Optimization, Intelligent Systems)

Mar 2026 – Expected Feb 2028

Universidad Industrial de Santander

Bucaramanga, Colombia

B.S. in Mechanical Engineering (Focus: Computational Modeling & Applied Mathematics)

Jul 2019 – Sep 2025

International Labour Organization (ILO)

Remote

Professional Diploma in Data Analytics

Aug 2023 – Nov 2023

TECHNICAL SKILLS

Languages: Python, Rust, C/C++, TypeScript, SQL, R, C# (.NET), Bash

AI, ML & Frameworks: PyTorch, CUDA, TensorRT, TensorFlow, ONNX Runtime, Hugging Face, Transformers, spaCy, NLTK, Scikit-learn, OpenCV, Pandas, NumPy

Core Domains: Natural Language Processing (NLP), Computer Vision, Multimodal Generative AI, Speech Systems (ASR/STT/TTS), Conversational AI

Distributed Systems & Databases: Apache Kafka, RabbitMQ, Actix-web, RESTful APIs, Microservices, MongoDB, MySQL, Microsoft SQL Server, PostgreSQL

DevOps & Cloud: Git, GitHub Actions, Docker, Kubernetes, CI/CD, Microsoft Azure, Google Cloud Platform (GCP), Linux